

Intra-Plane Coupling in the Multiple Scattering of Elastodynamic Waves in a Depth-Stratified Medium

Plane-Wave Propagators, Conditional Convergence, and Ewald Summation

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Abstract

The Green's tensor of a depth-stratified elastodynamic medium admits a partial-wave (plane-wave) expansion in which the depth-direction wavenumber of each constituent wave is fixed, through the dispersion relation, by the frequency, the local wavespeed of that partial wave, and the horizontal wavenumber. Used as an inter-site propagator for multiple-scattering sites, this representation is optimal when the sites lie at different depths: the evanescent spectrum supplies an exponential convergence factor whose rate grows with the depth separation. The same representation fails to converge when two sites share a depth, because that exponential damping disappears. This note records why, and then develops the standard cure for the intra-plane (equal-depth) coupling: Ewald summation of the Bloch-periodic lattice sum. We present the method in the language of analytic number theory, where it is the theta-splitting argument used to continue an Epstein zeta function, give the explicit real-space and reciprocal-space halves with their Gaussian convergence factors, and reduce the elastodynamic tensor problem to two scalar Helmholtz lattice sums by an exact operator identity. Finally we derive the SH and P/SV traction kernels, the displacement-traction partial-wave system whose continuity across interfaces and scatterer surfaces closes the multiple-scattering problem, and check them against the Rayleigh secular equation. A closing section applies the convergence theory to a space-filling grid of scatterers: the coupling splits into the same-depth near field, an interplane specular response carried by the Kennett recursion, and an interplane side-scattering term that the same evanescent scale confines to the nearest diagonal neighbours. Same-depth and interplane near fields then merge into a single first-shell real-space stencil over a global spectral propagator.

Companion series. *One of a companion series on elastic multiple scattering from cubic heterogeneities on a depth-stratified background, the T -matrix programme $T = T_0(I - G_0T_0)^{-1}$: (I) the stratified inter-site propagator, the same-depth convergence theory, the SH and P/SV traction kernels, interplane side scattering and the first-shell near stencil (`EwaldIntraPlanePropagator`); (II) the contrast-source volume integral equation and the CG-FFT / seismic grid T -matrix lineage (`ContrastSourceVIE`); (III) the directional partial-wave sweep matvec and the Krylov multiple-scattering solve (`DirectionalSweepSolver`); with the empirical companion `RQ1_Planar_Decomposition_Proposal`. This is note (I).*

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1 The stratified Green's tensor and its partial-wave expansion

Let the elastic properties of the medium (density ρ and the isotropic moduli, equivalently the compressional and shear speeds α and β) depend only on the depth coordinate z , and be invariant under translations in the horizontal plane. That horizontal homogeneity is what permits a Fourier transform over the horizontal coordinates $\boldsymbol{\rho} = (x, y)$ and time t , reducing the full three-dimensional problem to a family of one-dimensional (in depth) problems, one for each horizontal wavevector $\mathbf{k}_{\parallel} = (k_x, k_y)$ and frequency ω . The Green's tensor $G_{ij}(\mathbf{r}, \mathbf{r}'; t)$ is expanded over these plane waves, each carrying a factor $\exp[i(\mathbf{k}_{\parallel} \cdot \boldsymbol{\rho} - \omega t)]$.

Within any locally homogeneous region the field decomposes into upgoing and downgoing partial waves: compressional (P) and vertically polarised shear (SV), together with the horizontally polarised shear (SH) waves that decouple. For a partial wave of local speed c the depth-direction wavenumber follows from the dispersion relation

$$k_z(c) = \sqrt{\frac{\omega^2}{c^2} - k_x^2 - k_y^2}, \quad c \in \{\alpha, \beta\}, \quad \text{Im } k_z \geq 0. \quad (1)$$

The total wavevector has fixed magnitude ω/c ; the vertical component k_z is whatever remains after the horizontal part is removed. Two partial waves at the same ω and $\mathbf{k}_{\parallel}$ therefore carry *different* vertical wavenumbers,

$$k_z^{(\alpha)} = \sqrt{\omega^2/\alpha^2 - k_{\parallel}^2} \quad (\text{P}), \quad k_z^{(\beta)} = \sqrt{\omega^2/\beta^2 - k_{\parallel}^2} \quad (\text{S}), \quad (2)$$

with $k_{\parallel} = |\mathbf{k}_{\parallel}|$. In ray-parameter form, with horizontal slowness $p = k_{\parallel}/\omega$ and vertical slownesses $q_c = \sqrt{1/c^2 - p^2}$, one has $k_z = \omega q_c$.

The branch in (1) carries the essential physics. When $\omega^2/c^2 > k_{\parallel}^2$ the wave propagates obliquely (k_z real); when $\omega^2/c^2 < k_{\parallel}^2$ it is evanescent (k_z imaginary), decaying with depth. The evanescent branch is what carries head waves, post-critical reflections, and interface and surface modes. The whole stratification enters diagonally in $(\mathbf{k}_{\parallel}, \omega)$: reflection and transmission from the layering appear as coefficients multiplying the up- and downgoing partial waves, stitched across interfaces by continuity of displacement and traction.

2 The inter-site propagator and depth-separation convergence

For a homogeneous reference medium the scalar building block of the propagator is the outgoing Helmholtz kernel, and its plane-wave (Weyl, or Sommerfeld) representation is

$$\frac{e^{i\kappa r}}{4\pi r} = \frac{i}{8\pi^2} \int_{\mathbb{R}^2} \frac{d^2\mathbf{k}_{\parallel}}{k_z} e^{i(\mathbf{k}_{\parallel} \cdot \boldsymbol{\rho} + k_z |z|)}, \quad \kappa = \omega/c, \quad k_z = \sqrt{\kappa^2 - k_{\parallel}^2}, \quad \text{Im } k_z \geq 0. \quad (3)$$

The depth dependence sits entirely in $e^{ik_z|z-z'|}$. Past the radiative cone, $k_{\parallel} > \kappa$, we have $k_z = i\kappa_{\perp}$ with $\kappa_{\perp} = \sqrt{k_{\parallel}^2 - \kappa^2} > 0$, and that factor becomes a real exponential decay,

$$e^{ik_z|\Delta z|} = e^{-\kappa_{\perp}|\Delta z|}, \quad \Delta z = z - z'. \quad (4)$$

This is the only mechanism that tames the large- $k_{\parallel}$ tail of the spectral integral, or, for a periodic arrangement of sites, of the lattice sum. Examine that tail. At large $k_{\parallel}$, $1/k_z \sim 1/(ik_{\parallel})$, the two-dimensional measure contributes $k_{\parallel} dk_{\parallel}$, and the kernel contributes $e^{-k_{\parallel}|\Delta z|}$, so

$$\int_0^{\infty} k_{\parallel} dk_{\parallel} \frac{1}{k_{\parallel}} e^{-k_{\parallel}|\Delta z|} = \int_0^{\infty} dk_{\parallel} e^{-k_{\parallel}|\Delta z|} = \frac{1}{|\Delta z|}. \quad (5)$$

The radial integral is finite if and only if $\Delta z \neq 0$, and it converges *faster the larger* $|\Delta z|$ is. This is precisely why the partial-wave representation is the optimal inter-site propagator across depths: the stratification is already diagonal in $(\mathbf{k}_{\parallel}, \omega)$, and the evanescent spectrum self-truncates exponentially at a rate set by the depth offset. Two sites at different depths, even directly above one another, are joined by an absolutely and rapidly convergent sum.

3 Breakdown of the representation at equal depth

Set $\Delta z = 0$. The damping factor becomes $e^{-\kappa_{\perp} \cdot 0} = 1$, the high- $k_{\parallel}$ evanescent waves are no longer suppressed, and (5) collapses to $\int_0^{\infty} dk_{\parallel}$, which diverges. Only the horizontal phase $e^{i\mathbf{k}_{\parallel} \cdot \Delta \boldsymbol{\rho}}$ remains:

- **Distinct in-plane sites** ($\Delta \boldsymbol{\rho} \neq 0$, $\Delta z = 0$): the sum is only *conditionally* convergent. Its value depends on the order of summation, and as a lattice sum it is the classic non-convergent structure constant.
- **Coincident sites** ($\Delta \boldsymbol{\rho} = 0$, $\Delta z = 0$): the sum diverges outright, which is simply the near-field singularity of the Green's tensor appearing unmasked, since that singularity *is* the large- $k_{\parallel}$ evanescent content.

It is worth stressing that the physical propagator between two distinct equal-depth sites is perfectly finite. What fails is this particular representation, not the quantity it represents. The evanescent expansion converges to the correct value everywhere off the plane and loses convergence on it.

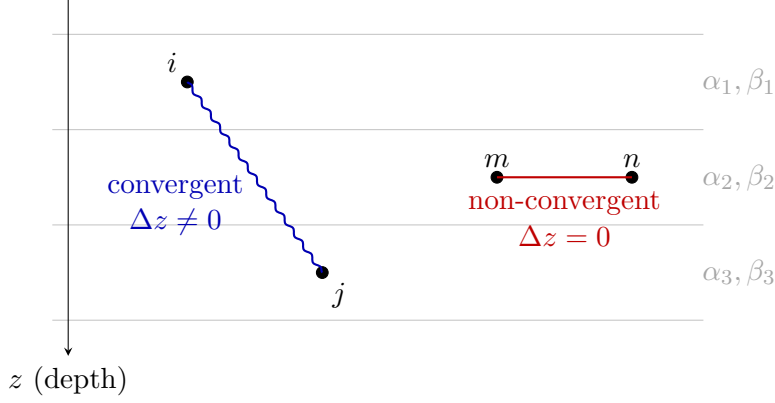


Figure 1: The plane-wave inter-site propagator. A vertical separation $\Delta z \neq 0$ supplies the evanescent damping $e^{-\kappa_{\perp}|\Delta z|}$ that makes the spectral sum converge (blue). Sites sharing a depth, $\Delta z = 0$, lose that factor and the representation ceases to converge (red).

This is the same split that layered multiple-scattering theory lives with, and the elastodynamic counterpart of the layer-KKR / LEED structure-constant problem of Korringa, Kohn and Rostoker, of Kambe, and of Pendry and Modinos: a plane-wave (reciprocal-space) propagator between sites in different planes, and a separate real-space construction, or an Ewald resummation, for sites within one plane.

4 Intra-plane coupling by Ewald summation

4.1 The object, in analytic terms

Fix a two-dimensional Bravais lattice $\Lambda \subset \mathbb{R}^2$ of scatterer positions in the plane, a Bloch character $\mathbf{k}_{\parallel}$, and a wavenumber $\kappa = \omega/c$. The intra-plane propagator is the quasi-periodic Helmholtz lattice sum

$$\mathcal{G}(\mathbf{r}) = \sum_{\mathbf{R} \in \Lambda} \frac{e^{i\kappa|\mathbf{r}-\mathbf{R}|}}{4\pi|\mathbf{r}-\mathbf{R}|} e^{i\mathbf{k}_{\parallel} \cdot \mathbf{R}}. \quad (6)$$

The two representations of §2–§3 are its two faces: (6) is the real-space form (terms $\sim 1/|\mathbf{R}|$, conditionally convergent), and its Poisson dual over the reciprocal lattice Λ^* is the plane-wave form (terms $\sim 1/k_{z,\mathbf{G}}$, convergent off the plane, divergent on it). On the plane $z = 0$ neither series converges absolutely. The task is purely analytic: assign $\mathcal{G}$ its correct finite value and compute it efficiently.

4.2 The Gaussian representation and the split

Write each term through the Gaussian (Mellin, or heat-kernel) representation

$$\frac{e^{i\kappa R}}{4\pi R} = \frac{1}{2\pi^{3/2}} \int_0^{\infty} \exp\left(-R^2\xi^2 + \frac{\kappa^2}{4\xi^2}\right) d\xi, \quad (7)$$

where the lower endpoint is made convergent by a slight rotation of the contour (for $\kappa \rightarrow 0$ this is exactly the Γ -function integral behind ζ). Now cut the ξ -integral at an arbitrary $\eta > 0$ and sum:

$$\mathcal{G} = \mathcal{G}^{(1)} + \mathcal{G}^{(2)}, \quad \mathcal{G}^{(1)} \leftrightarrow \int_{\eta}^{\infty}, \quad \mathcal{G}^{(2)} \leftrightarrow \int_0^{\eta}. \quad (8)$$

The upper piece keeps the strong Gaussian cutoff $e^{-R^2\xi^2}$ and is summed in real space; the lower piece carries the long-range tail and is Poisson-transformed to the dual lattice. Because the

Gaussian is self-dual under the Fourier transform, the dual lattice also sees a Gaussian, so both halves converge at Gaussian rate. The split point η is free and the total is exactly η -independent: that invariance is the practical numerical check, and η is tuned to balance the two sums, typically $\eta \approx \sqrt{\pi}/\sqrt{A}$ with A the area of the unit cell.

4.3 Real-space half

Carrying out $\int_\eta^\infty$ term by term yields error functions,

$$\mathcal{G}^{(1)}(\mathbf{r}) = \frac{1}{8\pi} \sum_{\mathbf{R} \in \Lambda} \frac{e^{i\mathbf{k}_\parallel \cdot \mathbf{R}}}{|\mathbf{r} - \mathbf{R}|} \sum_{s=\pm 1} e^{s i\kappa |\mathbf{r} - \mathbf{R}|} \operatorname{erfc}\left(|\mathbf{r} - \mathbf{R}| \eta + s \frac{i\kappa}{2\eta}\right) \quad (9)$$

For large $|\mathbf{R}|$ the argument of each erfc tends to $+\infty$, so the term decays like $e^{-\eta^2 |\mathbf{R}|^2}$. The real-space sum is absolutely and Gaussian-fast convergent.

4.4 Reciprocal-space half

Poisson summation of $\int_0^\eta$ over Λ (the calculation is in the appendix) gives a sum over Λ^* . Specialised to the plane $z = 0$, where the bare spectral sum failed, it is

$$\mathcal{G}^{(2)}(\boldsymbol{\rho}, 0) = \frac{i}{2A} \sum_{\mathbf{G} \in \Lambda^*} \frac{e^{i(\mathbf{k}_\parallel + \mathbf{G}) \cdot \boldsymbol{\rho}}}{k_{z,\mathbf{G}}} \operatorname{erfc}\left(\frac{k_{z,\mathbf{G}}}{2i\eta}\right), \quad k_{z,\mathbf{G}} = \sqrt{\kappa^2 - |\mathbf{k}_\parallel + \mathbf{G}|^2}, \quad \operatorname{Im} k_{z,\mathbf{G}} \geq 0. \quad (10)$$

For evanescent orders, $k_{z,\mathbf{G}} = i\sqrt{|\mathbf{k}_\parallel + \mathbf{G}|^2 - \kappa^2}$, the erfc argument is real and positive and the term decays like $e^{-(|\mathbf{k}_\parallel + \mathbf{G}|^2 - \kappa^2)/4\eta^2}$, Gaussian in $|\mathbf{G}|$. The finitely many propagating orders ($|\mathbf{k}_\parallel + \mathbf{G}| < \kappa$) give an erfc of imaginary argument and are evaluated directly. Letting $\eta \rightarrow \infty$ sends $\operatorname{erfc} \rightarrow 1$ and recovers the bare, divergent spectral sum $\frac{i}{2A} \sum_{\mathbf{G}} k_{z,\mathbf{G}}^{-1} e^{i(\mathbf{k}_\parallel + \mathbf{G}) \cdot \boldsymbol{\rho}}$; the factor $\operatorname{erfc}(k_{z,\mathbf{G}}/2i\eta)$ is exactly the regulator that the equal-depth configuration was missing.

Half	Summed over	Convergence factor	Decay rate
real-space, (9)	direct lattice Λ	$\operatorname{erfc}(\mathbf{R} \eta + \frac{i\kappa}{2\eta})$	$e^{-\eta^2 \mathbf{R} ^2}$
reciprocal-space, (10)	dual lattice Λ^*	$\operatorname{erfc}(k_{z,\mathbf{G}}/2i\eta)$	$e^{- \mathbf{k}_\parallel + \mathbf{G} ^2/4\eta^2}$

Table 1: The two halves of the Ewald-summed intra-plane propagator. Both are Gaussian-fast; their sum is independent of η .

4.5 Why a mathematician already knows this

Equations (7)–(10) are the theta-splitting argument that continues an Epstein zeta function and proves its functional equation. The Gaussian integral is the Mellin representation; cutting at η is splitting the Mellin integral at a chosen point; Poisson summation of the Gaussian is the Jacobi imaginary transformation $\theta(1/t) = t^{1/2}\theta(t)$. Read “Ewald parameter” as “the cut in the theta integral,” and read convergence at $z = 0$ as nothing more mysterious than the analytic continuation of $\sum_{\mathbf{R}} Q(\mathbf{R})^{-s}$ past its abscissa of convergence. The mechanism is the self-duality of the Gaussian under the Fourier transform: it is the unique regulator that tames a lattice and its dual at once.

4.6 A finite heterogeneity: zero-padding into a corner

Sections §4.1–§4.4 compute the *periodic* structure constant: the reciprocal sum over Λ^* is the field of an infinite array that tiles the plane. A real heterogeneity is not that. It is a finite lateral patch of cells embedded in the layered background, and against a finite patch the periodic reciprocal sum is the wrong operator, because it couples the patch to spurious periodic images of itself. The fast Fourier transform that evaluates the far half is itself intrinsically a *circular* convolution: it wraps a cell at one edge of the box onto a cell at the opposite edge.

The cure is zero-padding, and its geometry is a corner (Figure 2). Embed the patch, of lateral extent L , in a computational box of side at least $2L$, so that the scatterers occupy one corner and the remainder holds none. The circular convolution over the padded box, read back on the patch, then equals the *linear* (aperiodic) convolution exactly: every wraparound image falls in the empty padding, a gap of at least L from the real cells, and never reaches them. For a finite region the periodic structure constant of §4.4 is thereby replaced by the zero-padded transform of the patch in a corner.

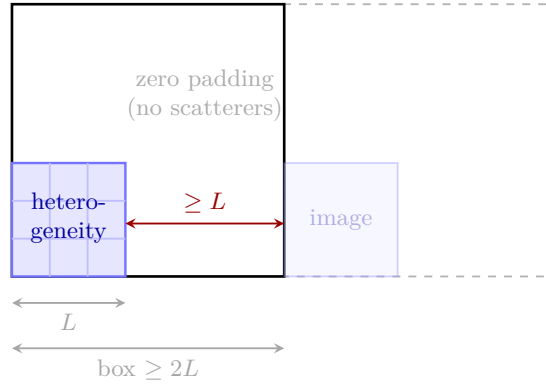


Figure 2: Zero-padded embedding of a finite heterogeneity for the aperiodic transform. The patch of lateral extent L sits in one corner of a box of side at least $2L$, the remainder carrying no scatterers. The circular convolution the FFT performs then equals the linear, aperiodic convolution on the patch, because the nearest periodic image (dashed) is held at least a gap L away in the empty padding and never reaches a real cell. Only the reciprocal half changes; the real-space near stencil is untouched.

Two points make this the natural update rather than a patch on the method. First, only the reciprocal (far) half changes: the real-space near half of §4.3, the first-shell stencil of §7.4, is local and periodicity-agnostic and is untouched. Second, because the patch is small and localised, the transform is taken over the box alone, while the infinite layered background outside it is carried analytically by the propagator, with no discretisation reaching to infinity and no absorbing boundary. The cost is a box doubled per lateral dimension, four times the area and transform work in the plane, in exchange for periodic images that contribute exactly zero. This is the exact counterpart of re-imposing periodicity through a large supercell with absorption: the padding removes the images without approximation.

5 Reducing the elastic tensor to scalar lattice sums

One does not need a separate dyadic theory. In a homogeneous reference medium the elastodynamic Green’s tensor is an exact, fixed second-order differential operator applied to two scalar Helmholtz kernels, one at the P-number $\kappa_P = \omega/\alpha$ and one at the S-number $\kappa_S = \omega/\beta$:

$$G_{ij}(\mathbf{r}) = \frac{1}{\rho\omega^2} \left[(\kappa_S^2 \delta_{ij} + \partial_i \partial_j) g_S(\mathbf{r}) - \partial_i \partial_j g_P(\mathbf{r}) \right], \quad g_c(\mathbf{r}) = \frac{e^{i\kappa_c r}}{4\pi r}. \quad (11)$$

This identity is exact, not asymptotic. In Fourier space the longitudinal and transverse projectors make the $1/k^2$ near-field pieces cancel identically, so (11) already contains the full near field. Because $\partial_i \partial_j$ acts on the field point while the lattice sum runs over source translations, the operator commutes through the sum. Hence the periodic elastodynamic structure constant is

$$\mathcal{G}_{ij} = \frac{1}{\rho\omega^2} \left[(\kappa_S^2 \delta_{ij} + \partial_i \partial_j) \mathcal{G}_{\kappa_S} - \partial_i \partial_j \mathcal{G}_{\kappa_P} \right], \quad (12)$$

where $\mathcal{G}_{\kappa_P}$ and $\mathcal{G}_{\kappa_S}$ are the scalar Ewald sums (9)–(10) evaluated at κ_P and κ_S . The recipe is then: Ewald-accelerate the two scalar sums, then apply the derivatives. On the reciprocal half the differentiation is algebraic: $\partial_j \mapsto i(\mathbf{k}_{\parallel} + \mathbf{G})_j$ in-plane, so it costs nothing and preserves Gaussian convergence; on the real half it pulls polynomials in front of the Gaussians, again preserving the rate.

The one caveat. $\partial_i \partial_j g_S$ carries a distributional $\delta^{(3)}$ contribution (the depolarisation, or L -dyadic, term) at coincidence. This bites only at the self-site $\mathbf{r} = \mathbf{0}$, which in multiple-scattering theory is handled separately by the on-site t -matrix. For genuine inter-site separations $\mathbf{r} = \mathbf{R}_i - \mathbf{R}_j \neq \mathbf{0}$, including all equal-depth pairs, the derivatives are classical and there is nothing to regularise beyond the lattice sum itself.

6 Traction kernels and closure of the multiple-scattering system

The structure constants of §4 propagate displacement between sites. To close the multiple-scattering system one also needs the *traction* carried by each partial wave, because the conditions that fix the single-scatterer response and the host interfaces (welded contact, free surface, fluid-solid contact) are stated in displacement and traction together. Rotate the horizontal axes so that $\mathbf{k}_{\parallel} = k \hat{\mathbf{k}}$ with $k = |\mathbf{k}_{\parallel}|$, and let $\hat{\phi} = \hat{\mathbf{z}} \times \hat{\mathbf{k}}$. Then P and SV motion lives in the sagittal plane $(\hat{\mathbf{k}}, \hat{\mathbf{z}})$ and SH along $\hat{\phi}$; a general azimuth is restored by the rotation $\mathbf{k}_{\parallel}/k$. Write the vertical wavenumbers $\nu_P = \sqrt{\kappa_P^2 - k^2}$ and $\nu_S = \sqrt{\kappa_S^2 - k^2}$ of (1), with $\text{Im } \nu \geq 0$.

6.1 The traction operator on a horizontal interface

With isotropic Hooke's law $\sigma_{ij} = \lambda \delta_{ij} \partial_k u_k + \mu(\partial_i u_j + \partial_j u_i)$, the traction on a plane $z = \text{const}$ is $\mathbf{t} = (\sigma_{\hat{k}z}, \sigma_{\phi z}, \sigma_{zz})$, with

$$\sigma_{\hat{k}z} = \mu(\partial_{\hat{k}} u_z + \partial_z u_{\hat{k}}), \quad \sigma_{\phi z} = \mu(\partial_{\phi} u_z + \partial_z u_{\phi}), \quad \sigma_{zz} = \lambda \nabla \cdot \mathbf{u} + 2\mu \partial_z u_z. \quad (13)$$

In the horizontal-wavenumber domain $\partial_{\hat{k}} \rightarrow ik$ and $\partial_{\phi} \rightarrow 0$, so the operator is algebraic in k and differential in z .

6.2 SH: the scalar traction kernel

The SH displacement $\mathbf{u} = u_{\phi}(z) \hat{\phi} e^{i\mathbf{k}_{\parallel} \cdot \mathbf{r}}$ is divergence-free, so $\sigma_{zz} = 0$ and the only surviving traction is along $\hat{\phi}$:

$$\boxed{t_{\phi} = \mu \partial_z u_{\phi}}, \quad u_{\phi} = D e^{i\nu_S z} + U e^{-i\nu_S z}, \quad t_{\phi} = i\mu\nu_S (D e^{i\nu_S z} - U e^{-i\nu_S z}). \quad (14)$$

The vertical SH impedance is $i\mu\nu_S$: each partial wave carries traction $\pm i\mu\nu_S$ per unit displacement (lower sign upgoing). Closure is immediate. A free surface ($t_{\phi} = 0$) gives $D = U$, reflection coefficient +1. A welded contact at z_0 (continuity of u_{ϕ} and t_{ϕ}), for incidence from medium 1, gives

$$R_{\text{SH}} = \frac{\mu_1 \nu_{S1} - \mu_2 \nu_{S2}}{\mu_1 \nu_{S1} + \mu_2 \nu_{S2}}, \quad T_{\text{SH}} = \frac{2\mu_1 \nu_{S1}}{\mu_1 \nu_{S1} + \mu_2 \nu_{S2}}. \quad (15)$$

The shear impedance $\mu\nu_S$ is the whole SH content, and is also the scalar Dirichlet-to-Neumann map $t_{\phi} = i\mu\nu_S u_{\phi}$ of a lower half-space.

6.3 P/SV: the coupled traction system

Collect the motion-stress vector $\mathbf{b} = (u_{\hat{k}}, u_z, t_{\hat{k}}, t_z)^\top$ with $t_{\hat{k}} = \sigma_{\hat{k}z}$, $t_z = \sigma_{zz}$ [13]. The equation of motion $\partial_j \sigma_{ij} + \rho \omega^2 u_i = 0$ with Hooke's law gives the first-order Thomson–Haskell system $\mathbf{b}'(z) = \mathbf{A} \mathbf{b}(z)$ [11, 12],

$$\mathbf{A} = \begin{pmatrix} 0 & -ik & \mu^{-1} & 0 \\ -\frac{i\lambda k}{\lambda + 2\mu} & 0 & 0 & \frac{1}{\lambda + 2\mu} \\ \frac{4\mu(\lambda + \mu)k^2}{\lambda + 2\mu} - \rho\omega^2 & 0 & 0 & -\frac{i\lambda k}{\lambda + 2\mu} \\ 0 & -\rho\omega^2 & -ik & 0 \end{pmatrix}, \quad (16)$$

whose eigenvalues are $\pm i\nu_P, \pm i\nu_S$. The eigenvectors are the partial-wave displacement-traction kernels. With displacement conventions $P \propto (k, \pm\nu_P)$ (longitudinal) and $SV \propto (\pm\nu_S, -k)$ (transverse, perpendicular to the P direction), upper sign downgoing, and writing $C \equiv \kappa_S^2 - 2k^2$ (so $\mu C = \rho\omega^2 - 2\mu k^2$), the four columns assemble into the fundamental matrix

$$\mathbf{D} = \begin{pmatrix} k & k & \nu_S & -\nu_S \\ \nu_P & -\nu_P & -k & -k \\ 2i\mu k \nu_P & -2i\mu k \nu_P & i\mu C & i\mu C \\ i\mu C & i\mu C & -2i\mu k \nu_S & 2i\mu k \nu_S \end{pmatrix}, \quad \mathbf{b}(z) = \mathbf{D} \text{diag}(e^{i\nu_P z}, e^{-i\nu_P z}, e^{i\nu_S z}, e^{-i\nu_S z}) \mathbf{c}, \quad (17)$$

with columns ordered $(P\downarrow, P\uparrow, SV\downarrow, SV\uparrow)$ and $\mathbf{c}$ the partial-wave amplitudes. The upper two rows are the displacement kernels; the lower two are the traction kernels. The recurring entries $\mu C = \rho\omega^2 - 2\mu k^2$ and $2\mu k \nu$ are the P/SV coupling kernels, and they are what convert P into SV at every interface. (Columns may be flux-normalised, for instance dividing the P columns by κ_P and the SV columns by κ_S , without changing this structure.)

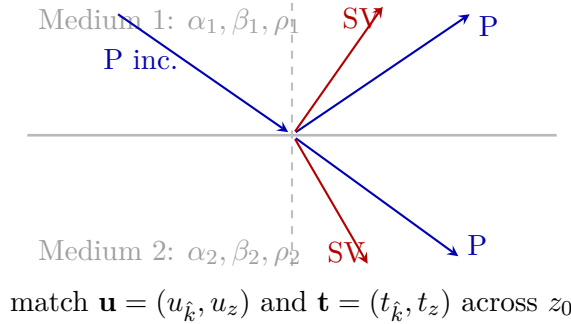


Figure 3: Mode conversion at a welded horizontal interface: an incident P wave produces reflected and transmitted P and SV (SH decouples). The displacement kernels (upper two rows of $\mathbf{D}$) and traction kernels (lower two rows) are matched across z_0 ; the coupling entries $\mu C = \rho\omega^2 - 2\mu k^2$ and $2\mu k \nu$ drive the conversion.

6.4 Surface impedance and the Rayleigh check

Write $\mathbf{D} = \begin{bmatrix} \mathbf{D}_u \\ \mathbf{D}_t \end{bmatrix}$, with $\mathbf{D}_u$ the displacement rows and $\mathbf{D}_t$ the traction rows, and let the superscript $\downarrow$ select the two columns that decay into a lower half-space ($P\downarrow, SV\downarrow$). Eliminating those amplitudes gives the Dirichlet-to-Neumann (surface-impedance) map $\mathbf{t} = \mathbf{Z} \mathbf{u}$,

$$\mathbf{Z} = \mathbf{D}_t^\downarrow (\mathbf{D}_u^\downarrow)^{-1} = \frac{i\mu}{k^2 + \nu_P \nu_S} \begin{pmatrix} \kappa_S^2 \nu_P & k B \\ -k B & \kappa_S^2 \nu_S \end{pmatrix}, \quad B = 2k^2 + 2\nu_P \nu_S - \kappa_S^2, \quad (18)$$

with $\mathbf{D}_u^\downarrow = \begin{pmatrix} k & \nu_S \\ \nu_P & -k \end{pmatrix}$ and $\mathbf{D}_t^\downarrow = \begin{pmatrix} 2i\mu k\nu_P & i\mu C \\ i\mu C & -2i\mu k\nu_S \end{pmatrix}$. This $\mathbf{Z}(\mathbf{k}_\parallel, \omega)$ is the traction kernel in its most usable form: the traction that any prescribed surface displacement must balance. A traction-free surface of the half-space requires $\det \mathbf{D}_t^\downarrow = 0$, that is

$$(2k^2 - \kappa_S^2)^2 + 4k^2\nu_P\nu_S = 0 \iff (2k^2 - \kappa_S^2)^2 = 4k^2\sqrt{k^2 - \kappa_P^2}\sqrt{k^2 - \kappa_S^2}, \quad (19)$$

the Rayleigh secular function, whose root is the Rayleigh wave. The analogous condition at a welded solid-solid contact yields the Stoneley wave. Recovering the Rayleigh equation is a complete check on the P/SV traction kernels.

6.5 Closing the system

The single-scatterer response and the host interfaces now live in one basis.

- **Free surface and welded interface.** A free surface imposes $\mathbf{D}_t \mathbf{c} = \mathbf{0}$, giving the free-surface reflection matrix $\mathbf{R}_F = -(\mathbf{D}_t^{\text{ref}})^{-1} \mathbf{D}_t^{\text{inc}}$. A welded interface imposes continuity of $\mathbf{D}_u \mathbf{c}$ and $\mathbf{D}_t \mathbf{c}$ across z_0 , giving the interface reflection and transmission matrices $\mathbf{R}, \mathbf{T}$; SH closes the same way through $i\mu\nu_S$.
- **Single-scatterer T-matrix.** On a scatterer boundary the condition is written through the same displacement and traction kernels: traction-free for a cavity, displacement-free for a rigid body, continuity of both for a welded inclusion. Projected onto the partial-wave basis, this yields the scatterer T-matrix $\mathbf{T}_j$.
- **The closed system.** With the inter-site structure constants $\mathbf{G}_{ij}$ of §4 (intra-plane pieces Ewald-summed) carrying displacement and, through (17)–(18), traction between sites, the self-consistent multiple-scattering equations are

$$\mathbf{c}_i^{\text{exc}} = \mathbf{c}_i^0 + \sum_{j \neq i} \mathbf{G}_{ij} \mathbf{T}_j \mathbf{c}_j^{\text{exc}}, \quad i = 1, \dots, N, \quad (20)$$

or, for a stratified host, the equivalent recursive reflection-transmission addition. The traction kernels enter twice: in building each $\mathbf{T}_j$, and in matching the host's free surface and welded interfaces.

7 Interplane side scattering and the nearest-neighbour correction

On a space-filling grid of scatterers on the layered background, the convergence theory of §2–§4 dictates a three-way split of the multiple-scattering coupling, the off-diagonal blocks of the Foldy–Lax system [14, 15],

$$G_0^{\text{used}} = \underbrace{G_0^{\text{intra}}}_{\Delta z=0} + \underbrace{G_0^{\text{Kennett}}}_{\text{interplane specular}} + \underbrace{G_0^{\text{side}}}_{\Delta z \neq 0, \Delta \rho \neq 0}. \quad (21)$$

The same-depth coupling G_0^{intra} is the spatial near field of §3, regularised at coincidence by the cubic-cell T-matrix of §5. The interplane specular response is carried by the Kennett reflectivity recursion, exact and unconditionally stable for the layered, laterally homogeneous part. What the Kennett recursion cannot carry is the lateral momentum transfer between planes, the scattering $\mathbf{k}_\parallel \rightarrow \mathbf{k}'_\parallel$ produced by the lateral heterogeneity of each plane. That is the interplane side scattering G_0^{side} , the one term the two clean pieces leave uncovered.

7.1 The side-scattering kernel

Between a cell in one plane and a cell in the adjacent plane at vertical offset h and lateral offset $\Delta\boldsymbol{\rho}$, the direct coupling is the elastic tensor built on $e^{i\kappa_c r}/4\pi r$ with $r = \sqrt{|\Delta\boldsymbol{\rho}|^2 + h^2}$, carrying the Kupradze near-field hierarchy

$$G_{ij}^{\text{side}} \sim \frac{A_{ij}(\hat{\mathbf{r}})}{r} e^{i\kappa_c r} + \frac{B_{ij}(\hat{\mathbf{r}})}{r^2} + \frac{C_{ij}(\hat{\mathbf{r}})}{r^3}, \quad (22)$$

the radiation ($1/r$), intermediate ($1/r^2$), and static near-field ($1/r^3$) terms. The $1/r^3$ term is excited by the modulus (strain) contrast and dominates at small ka ; the $1/r$ term is the displacement coupling of the density contrast. Kennett already carries the smooth, laterally averaged part of (22); the side correction supplies only the fluctuation about it.

7.2 Two ranges, both set by the plane spacing

That fluctuation is short ranged in two senses, and the same scale h governs both.

Vertically, only adjacent planes. The interplane coupling at vertical offset Nh carries the evanescent damping $e^{-\kappa_\perp Nh}$ of (5) and grows more specular as its lateral structure washes out with N . Distant planes are therefore both damped and well represented by Kennett; only $N = 1$, and marginally $N = 2$, needs a side correction.

Laterally, only the first shell. At fixed h the separation is $r = \sqrt{|\Delta\boldsymbol{\rho}|^2 + h^2}$, so the near-field terms turn over at $|\Delta\boldsymbol{\rho}| \sim h$. The dominant $1/r^3$ contribution falls between the first diagonal shell ($r = \sqrt{2}h$ at an edge, $\sqrt{3}h$ at a corner) and the second ($r \approx \sqrt{5}h$) by a factor $(\sqrt{5}/\sqrt{2})^3 \approx 4$, so the nearest diagonal neighbours dominate the strain coupling. Equivalently, the evanescent spectrum that carries the non-specular field is cut at $k_\parallel \sim 1/h$, whose real-space dual is a kernel of width $\sim h$. For space-filling cells h is one pitch, and that width is one cell.

The two ranges coincide because they are the same length h : the plane spacing that makes the Kennett recursion converge *between* planes is the spacing that confines the side scattering *across* them. For space-filling cells the correction is the single shell of diagonal neighbours, the 4 edge-sharing cells (lateral offset one pitch) and 4 corner-sharing cells (offset $\sqrt{2}$ pitch) in each adjacent plane.

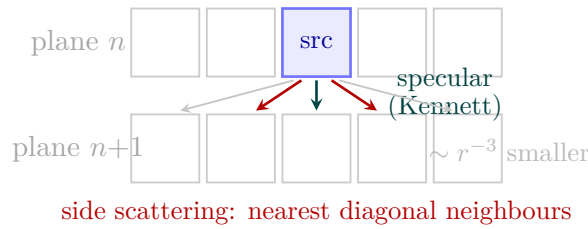


Figure 4: Interplane side scattering on a space-filling grid (cross-section). Kennett carries the specular vertical coupling; the lateral heterogeneity produces side scattering to laterally offset cells. The near-field hierarchy confines the missed coupling to the nearest diagonal shell (offset $\sim h$, one pitch), the second shell being smaller by the $1/r^3$ near field. The directly-below cell is specular and stays in Kennett, so the sparse correction takes only the diagonal neighbours.

7.3 The sparse correction and its residual

The side term is then a sparse, banded operator coupling each cell to those 8 diagonal neighbours per adjacent plane, evaluated as the volume-averaged spatial propagator and added to the

Kennett recursion. The cost is $O(N)$ with a small constant, and the directly-below cell is specular and remains in Kennett, so there is no double count. This is the augmented planar scheme: not pure planar, which drops G_0^{side} , and not full three-dimensional near field, which pays for all of it, but Kennett plus one shell.

The truncation error of the one-shell correction is the second diagonal shell, governed by the ratio of the slow $1/r$ (density) coupling to the fast $1/r^3$ (modulus) coupling and by ka . In the sub-resonant, modulus-dominated regime the $1/r^3$ decay gives a per-shell factor ~ 4 and the correction converges at the first shell; as $ka \rightarrow 1$ the $1/r$ radiation term grows and the range lifts. This is the falsifiable content of the nearest-neighbour hypothesis: a shell-convergence sweep (0 shells = pure planar, 1 = nearest, 2, ...) should plateau after the first shell through the Rayleigh-to-moderate regime, the plateau lifting toward resonance. An empirical companion study measures exactly that plateau and locates the ka at which a second shell becomes necessary.

7.4 One stencil for the whole near field

The range argument of §7.2 is not special to the interplane term. At $\Delta z = 0$ the kernel (22) holds with $r = |\Delta \boldsymbol{\rho}|$, and its $1/r^3$ part is equally short ranged, so the same-depth near coupling is nearest-neighbour dominated as well. The precise statement, as for the interplane term, is that it is the spatial near *correction* that is local, not the full same-depth coupling: the long $1/r$ radiation tail of §2–§3 does not truncate, but it is carried by the spectral half, the Ewald-reciprocal sum of §4, exactly as the interplane smooth part is carried by Kennett. The two near corrections differ only in which cheap carrier absorbs their far field.

The entire near field, same-depth and interplane side together, is then a single sparse real-space stencil over the first three-dimensional shell of the grid (Table 2).

Cells	Count	$(\Delta z, \Delta \boldsymbol{\rho})$	Role	Carried by
self	1	(0, 0)	on-site T_0	cubic-cell T-matrix (§5)
vertical faces	2	(h , 0)	specular	Kennett recursion
in-plane ring	8	(0, $\{h, \sqrt{2}h\}$)	same-depth near	real-space stencil
adjacent diagonals	16	(h , $\{h, \sqrt{2}h\}$)	interplane side	real-space stencil
remainder	—	$r \gtrsim \sqrt{3}h$	smooth far field	spectral propagator

Table 2: The first three-dimensional shell of the cubic grid and its roles. The self cell is the solved Eshelby/Galerkin T_0 ; the two vertical faces are specular and ride the Kennett recursion; the 24 in-plane and diagonal neighbours are the sparse real-space near correction; everything beyond is the smooth far field carried by the spectral layered propagator (the Ewald-reciprocal sum of §4 with the Kennett z -kernel).

Everything collapses to a two-piece operator,

$$G_0 = \underbrace{G_0^{\text{near}}}_{\text{24-neighbour real-space stencil}} + \underbrace{G_0^{\text{far}}}_{\text{global spectral layered propagator}}, \quad (23)$$

the sparse near correction (volume-averaged, so the strongest coupling, the in-plane face-sharing neighbour, is represented faithfully) plus the smooth far field carried by the Ewald-reciprocal / FFT sum with the Kennett reflectivity z -kernel. This is the precorrected-FFT structure of layered-medium integral equations, and the Ewald split of §4 is its periodic realisation. The whole development reduces to one rule: solve the first three-dimensional shell in real space, and everything beyond it in the spectral layered propagator.

The caveats are uniform across the stencil. It is a $1/r^3$, modulus-dominated, small- ka statement; the slow $1/r$ density coupling rides the spectral half; and as $ka \rightarrow 1$ the radiation

term grows and the shell thickens, at which point the cubes are subdivided and the scale separation is restored at the finer grid.

8 Adjacent routes and lineage

- **Limiting absorption** is the conceptually cleanest regulator: give ω a positive imaginary part so every $e^{i\kappa R}/R$ decays and the real-space sum converges absolutely, then continue to real ω . Ewald is how that continuation is realised without the cost of literal damping or a slow real-space sum.
- **Singularity subtraction**: subtract the static ($\kappa = 0$) lattice sum, whose continuation is a known Epstein or Kambe object, and Ewald only the smoother remainder. This is useful near lattice resonances, where $k_{z,\mathbf{G}} \rightarrow 0$ for some order and that term needs care.
- **Lineage and references**. Ewald [1] for the original split; Kambe [3] and Ham and Segall [2] for the angular-momentum lattice sums of layer-KKR; Linton [4] for a mathematician-facing survey of Helmholtz lattice sums. For the elastodynamic layer-multiple-scattering construction specifically, Psarobas, Stefanou and Modinos [5] and Sainidou, Stefanou, Psarobas and Modinos [6] carry out exactly the P/S scalar reduction of (11). The cleanest write-ups of the dyadic-Ewald bookkeeping live in the electromagnetic periodic-Green's-function literature, Capolino, Wilton and Johnson [7], and transfer verbatim once the substitution g_P, g_S is made. For the elastodynamic and seismological background, Aki and Richards [8], Kennett [9], and Ben-Menahem and Singh [10].

A The reciprocal half by Poisson summation

With $f(\boldsymbol{\rho}) = e^{-|\boldsymbol{\rho}|^2 \xi^2}$ the two-dimensional Poisson summation formula reads

$$\sum_{\mathbf{R} \in \Lambda} f(\boldsymbol{\rho} - \mathbf{R}) e^{i\mathbf{k}_{\parallel} \cdot \mathbf{R}} = \frac{1}{A} \sum_{\mathbf{G} \in \Lambda^*} \hat{f}(\mathbf{k}_{\parallel} + \mathbf{G}) e^{i(\mathbf{k}_{\parallel} + \mathbf{G}) \cdot \boldsymbol{\rho}}, \quad \hat{f}(\mathbf{k}) = \frac{\pi}{\xi^2} e^{-|\mathbf{k}|^2 / 4\xi^2}. \quad (24)$$

Insert this into $\mathcal{G}^{(2)} = \frac{1}{2\pi^{3/2}} \sum_{\mathbf{R}} e^{i\mathbf{k}_{\parallel} \cdot \mathbf{R}} \int_0^\eta e^{-|\mathbf{r} - \mathbf{R}|^2 \xi^2 + \kappa^2 / 4\xi^2} d\xi$, use $|\mathbf{r} - \mathbf{R}|^2 = |\boldsymbol{\rho} - \mathbf{R}|^2 + z^2$ and $|\mathbf{k}_{\parallel} + \mathbf{G}|^2 - \kappa^2 = -k_{z,\mathbf{G}}^2$, and substitute $\xi \mapsto 1/\xi$ in the radial integral. At $z = 0$ the integral evaluates to $\int_{1/\eta}^\infty e^{k_{z,\mathbf{G}}^2 w^2 / 4} dw = \frac{i\sqrt{\pi}}{k_{z,\mathbf{G}}} \operatorname{erfc}(k_{z,\mathbf{G}} / 2i\eta)$, which delivers (10). As a by-product, $\eta \rightarrow \infty$ returns the exact spectral representation of the periodic Green's function,

$$\mathcal{G}(\boldsymbol{\rho}, z) = \frac{i}{2A} \sum_{\mathbf{G} \in \Lambda^*} \frac{e^{i(\mathbf{k}_{\parallel} + \mathbf{G}) \cdot \boldsymbol{\rho} + i k_{z,\mathbf{G}} |z|}}{k_{z,\mathbf{G}}}, \quad (25)$$

whose $z = 0$ value is the conditionally convergent / divergent sum that motivated the whole construction.

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